HW4

1. In lower dimensions, like 2D or 3D, the volume of a ball is spread out relatively evenly across the ball. However, as dimension increases, the volume of the ball starts to concentrate near its surface. This happens because the volumes of a high-dimensional ball grows increasingly fast as one moves away from the center toward the surface.

In higher dimensions, the surface area of the ball (which is part of the sphere) becomes a larger and larger proportion of the total volume. This means that a significant part of the ball's volume is located in a thin layer close to its surface.

Mathematically, as the dimension d increases, the volume of the unit ball B^d grows at first, but then starts to shrink, concentrating near the boundary S^{d-1}.

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2.

① X is drawn from a multivariate normal distribution, $x_1, \cdots x_d$ are i.i.d and $x_i \sim N(0, \sigma^2)$.

$E[X_1^2 + X_2^2 + \cdots + X_d^2] = \sum_{i=1}^{d} (X^{(i)})^2$ follows a chi-squared distribution. Thus, $E[X_1^2 + X_2^2 + \cdots + X_d^2] = d\sigma^2 = 1$

$\Rightarrow \sigma^* = \frac{1}{\sqrt{d}}$

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3. Proof

$P(X_0=x_0, X_1=x_1, \dots X_n=x_n) = \mu(x_0)\cdot p(x_0,x_1)\cdot p(x_1,x_2)\cdots p(x_{n-1},x_n)$

for all $n=1,2,\dots$ and $x_0, x_1, \dots x_n \in \mathcal{X}$.

$P(X_0=x_0, X_1=x_1, \dots X_n=x_n)$

$= P(X_0=x_0) \times P(X_1=x_1 \mid X_0=x_0) \times \cdots \times P(X_{n-1}=x_{n-1} \mid X_{n-2}=x_{n-2}, \dots, X_0=x_0)$

$\times P(X_n=x_n \mid X_{n-1}=x_{n-1}, \dots X_0=x_0)$ Conditional Probability

$= P(X_0=x_0) \times P(X_1=x_1 \mid X_0=x_0) \times \cdots \times P(X_{n-1}=x_{n-1} \mid X_{n-2}=x_{n-2}) \times P(X_n=x_n \mid X_{n-1}=x_{n-1})$

Markov Property

$= \mu(x_0)\cdot p(x_0,x_1)\cdot p(x_1,x_2)\cdots p(x_{n-1},x_n)$ Homogeneity